



IIT Gandhinagar

PG Diploma in AI-Driven Cloud based Software Development

Curriculum and Learning Journey





Credit Structure (40 Credits)

Component	Credits	Hours
Theory (Lectures)	17	170 hrs
Practical (Labs + Capstone)	16	320 hrs
Competitive Coding & Ideas Lab	7	140 hrs
TOTAL	40	630 hrs

Credit Calculation: 10 teaching hours = 1 credit OR 20 practical hours = 1 credit

20-Week Learning Journey

Phase	Timeline	Subjects & Topics	Job Skills Unlocked
Foundation	Weeks 1-6	Python/Java, DSA, System Design, Git, Testing, CI/CD	Backend development, API design, database modeling, clean code practices
Specialization	Weeks 7-14	Full-Stack (React, Node.js), Backend Engineering, AI Integration, GenAI Tools	Full-stack apps, REST/GraphQL APIs, LLM integration, AI-powered features
Midterm	Week 15	Comprehensive technical assessment	Mid-program evaluation, interview readiness check
Capstone	Weeks 16-19	Microservices, Cloud (AWS/GCP), Docker, Kubernetes, DevOps, Production Deployment	Scalable architecture, cloud deployment, CI/CD pipelines, monitoring
Final Exam	Between Week 15-20	System design interviews, coding challenges, architectural discussions	Interview-ready for SWE, Full-Stack, Backend, DevOps Engineer roles



Month-by-Month: Subjects, Tools & Job Skills

Month 1 (Weeks 1-6): Engineering Toolkit

- Subjects: Python/Java Programming (6cr), DSA (5cr), System Design (4cr), Git & Testing (2cr)
- Tools: Python/Java, Git, PostgreSQL, Redis, Docker basics, pytest/JUnit, CI/CD fundamentals
- Job Skills: Clean code, SOLID principles, API design, database modeling, testing strategies

Month 2 (Weeks 7-10): Full-Stack Specialization

- Subjects: Full-Stack Development (7cr), Backend Engineering (5cr), AI Integration (4cr), GenAI Tools (2cr)
- Tools: React, Next.js, Node.js, Express, FastAPI, OpenAI/Claude APIs, LangChain, MongoDB, GraphQL
- Job Skills: Full-stack apps, REST/GraphQL APIs, LLM integration, real-time features, authentication

Months 3-4 (Weeks 11-17): Capstone Project

- Capstone (9cr): End-to-end SaaS application with microservices, cloud deployment, CI/CD, monitoring
- Tools: AWS/GCP/Azure, Docker, Kubernetes, Terraform, GitHub Actions, Prometheus, Grafana, DataDog
- Job Skills: Microservices architecture, cloud-native development, DevOps practices, scalability, observability

Month 5 (Weeks 18-20): Interview Mastery

- Focus: 300+ DSA problems, system design practice, 15+ mock interviews, behavioral prep
- Tools: LeetCode, HackerRank, system design platforms, mock interview sessions
- Job Skills: DSA mastery, system design (scale to millions), architectural trade-offs, behavioral stories

Throughout 20 Weeks: Competitive Coding & Ideas Lab (7cr)

- Daily DSA practice, code optimization, system design discussions, open-source contributions



Extra Value Pack

Full-Stack Development

- Frontend: React, Next.js, TypeScript, Tailwind CSS, state management (Redux, Zustand)
- Backend: Node.js, Express, FastAPI, Django, REST & GraphQL APIs

AI Integration & GenAI

- OpenAI, Claude, Gemini APIs, LangChain, vector databases, RAG implementation, AI agents

Cloud & DevOps

- AWS/GCP/Azure, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Jenkins), monitoring, logging

Databases & Caching

- PostgreSQL, MongoDB, Redis, database optimization, caching strategies, connection pooling

Agile for Engineering Teams

- Sprint planning, code reviews, technical documentation, architectural decision records (ADRs)

Professional Development

- 2 clubs: Entrepreneurs Club + AI Innovation Club (optional, Mon/Wed/Fri evenings). Coding tournaments & tech talks



Daily Learning Structure

Time (24-hour)	Activity
07:45 - 09:15	Breakfast at IIT-GN Campus
09:15 - 10:45	Session 1A - Core Concepts: Instructor live-codes applications, system design, architecture patterns with real-time Q&A
11:00 - 12:45	Session 1B - Hands-On Labs: Build features, APIs, deploy services. Pair programming, live debugging, code reviews
12:45 - 14:00	Lunch at IIT-GN Campus + Squad Sync: Discuss code, share solutions, plan afternoon builds
14:00 - 15:15	Session 2A - Production Patterns: Real-world architecture, scalability, security, performance optimization, industry practices
15:30 - 17:00	Session 2B - Advanced Implementation: Microservices, cloud deployment, mini-projects, peer reviews
17:00 - 17:30	Break + Snacks at IIT-GN Campus
17:30 - 18:00	Daily Standup: What shipped? What broke? Blockers? 2-min feature demos. Commit to GitHub
18:00 - 19:30	DSA Coding Hour: LeetCode battles, algorithm practice, squad competitions, daily leaderboard, interview patterns
19:30 onwards	Dinner at IIT-GN Campus, then hostel
Saturday	Weekly assessment + system design review + mock SWE interviews + coding competitions + club activities (optional)

Note: All meals (breakfast, lunch, snacks, dinner) are at IIT Gandhinagar campus mess